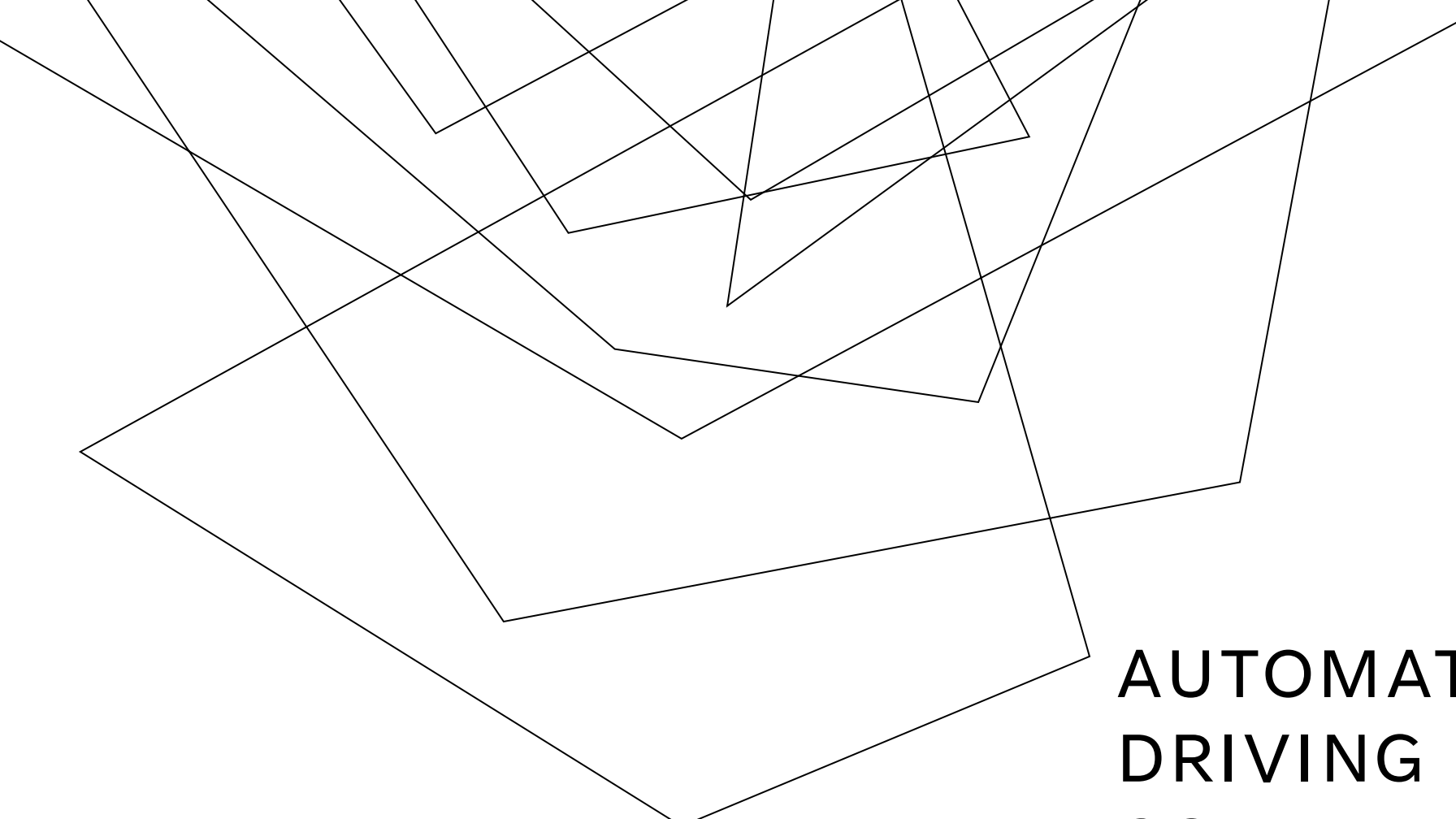


A series of thin, black, overlapping lines forming various geometric shapes like triangles and polygons, creating a complex, abstract pattern in the upper left portion of the slide.

AUTOMATIC DRIVING COMPUTING

5. PERCEPTION (3)





지난 시간 복습

OBJECT DETECTION (1)

1. Perception – overview
2. Object localization
3. Object detection
4. R-CNN





목차

목차

1. R-CNN inference

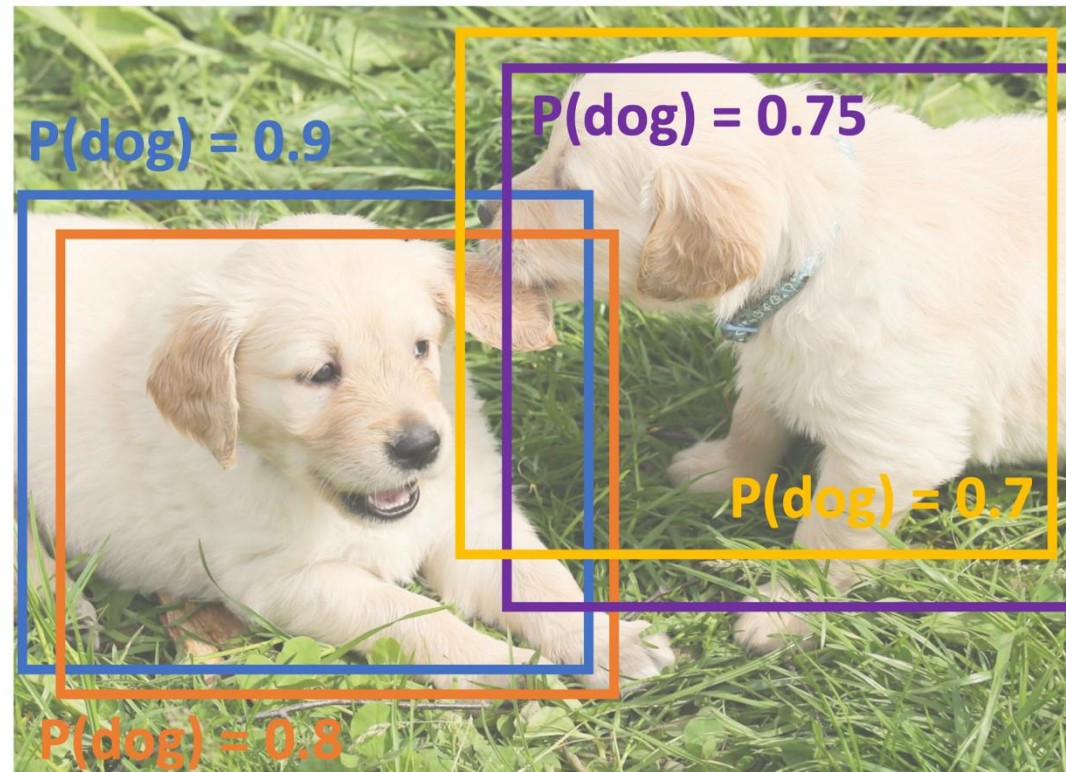
2. Fast R-CNN



R-CNN INFERENCE

모델 학습 끝난 후 - 하나의 물체에 대해 proposal 여러 개가 출력될 수 있음

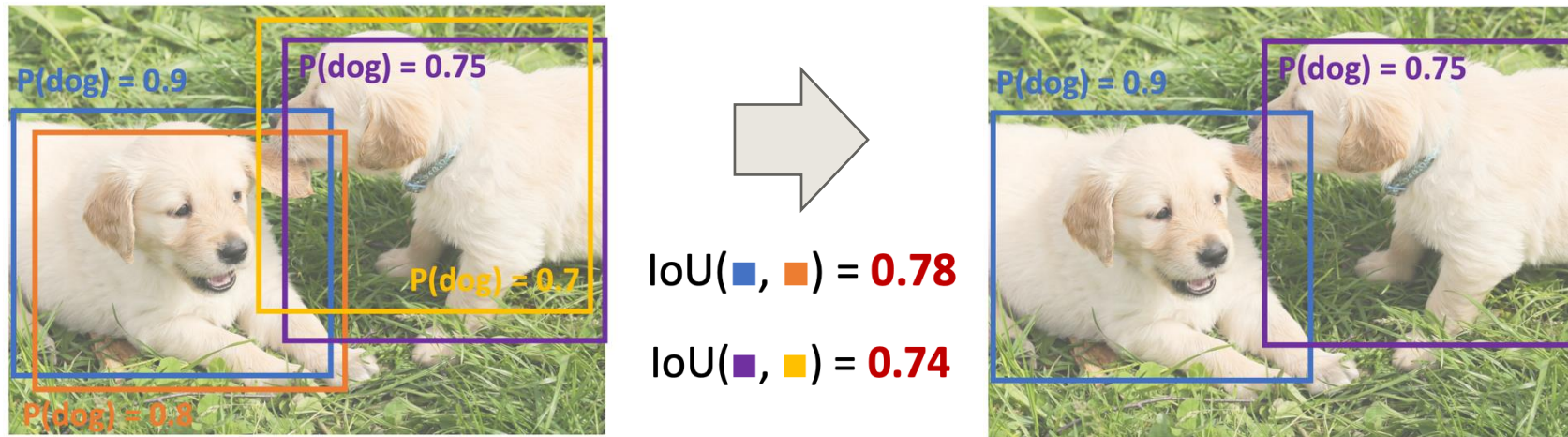
Cf) proposal의 category : argmax로 결정



R-CNN INFERENCE

Solution) Non-maximum suppression (NMS) 통한 후처리

- 1) 전체 proposal box를 confidence 내림차순으로 정렬
- 2) Confidence 높은 box 순서대로, 자신과 많이 겹치는 ($\text{IoU} > 0.7$) box 소거



Cf) 동일한 category 예측한 prediction들 사이에서만 소거 진행

R-CNN INFERENCE

After NMS, 모델의 성능 평가

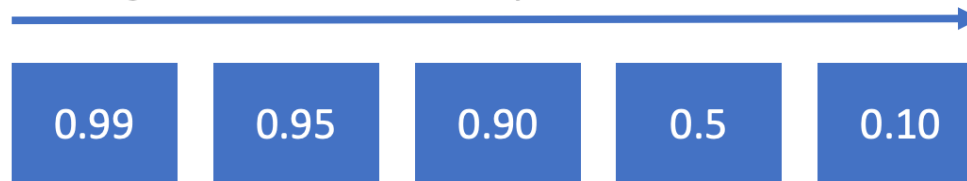
Metric : Average Precision (AP)

Q) 왜 Accuracy로 평가하지 않을까?

Category마다 독립적으로 평가

Cf) Dog category

All dog detections sorted by score

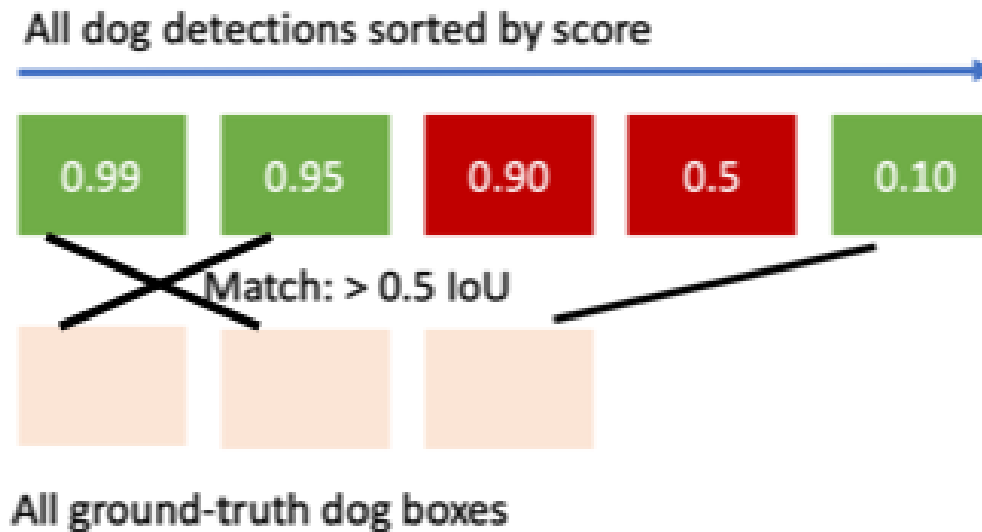


All ground-truth dog boxes



R-CNN INFERENCE

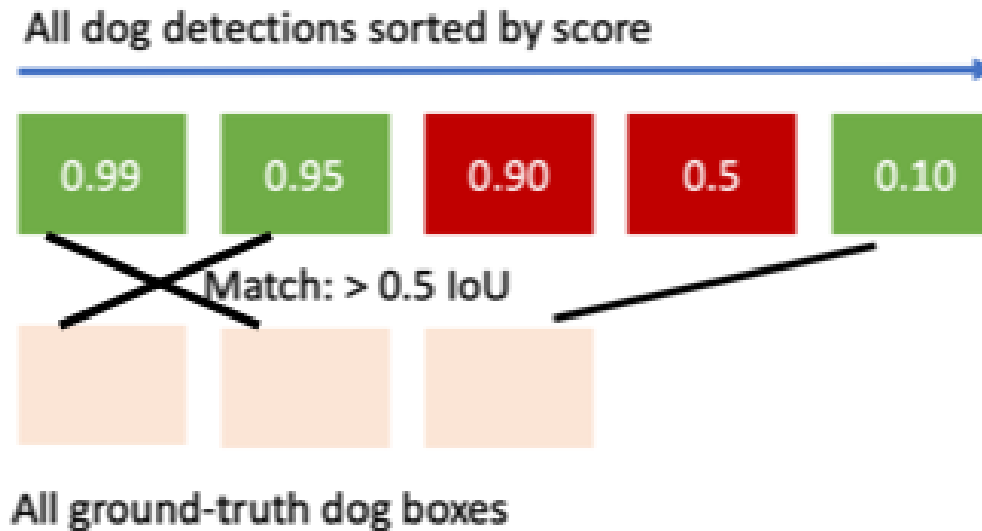
왼쪽에서 오른쪽 순서로 각 detection prediction에 대해서
Ground-truth box와 $\text{IoU} > 0.5$ 으로 매칭이 되면 Positive,
매칭이 안되면 Negative box로 지정



R-CNN INFERENCE

왼쪽에서 오른쪽 순서로 각 detection prediction에 대해서
Ground-truth box와 $\text{IoU} > 0.5$ 으로 매칭이 되면 Positive,
매칭이 안되면 Negative box로 지정

Q) Confidence threshold를 어떻게 설정?



R-CNN INFERENCE

“Precision-Recall tradeoff”

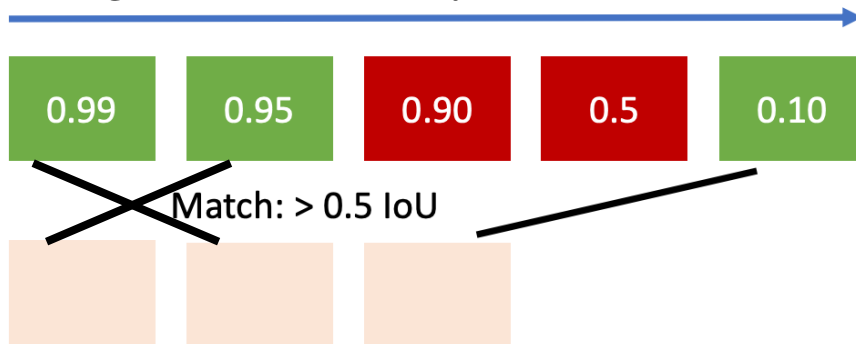
Confidence threshold 높으면 미탐 발생 → Recall (검출율) 감소

Confidence threshold 높으면 오탐 발생 → Precision (정확도) 감소

Ex1) threshold 0.95 → Precision : , Recall :

Ex2) threshold 0.1 → Precision : , Recall :

All dog detections sorted by score



All ground-truth dog boxes

실제 상황 (ground truth)	예측 결과 (predict result)	
	Positive	Negative
Positive	TP(true positive) 옳은 검출	FN(false negative) 검출되어야 할 것이 검출되지 않았음
Negative	FP(false positive) 틀린 검출	TN(true negative) 검출되지 말아야 할 것이 검출되지 않았음

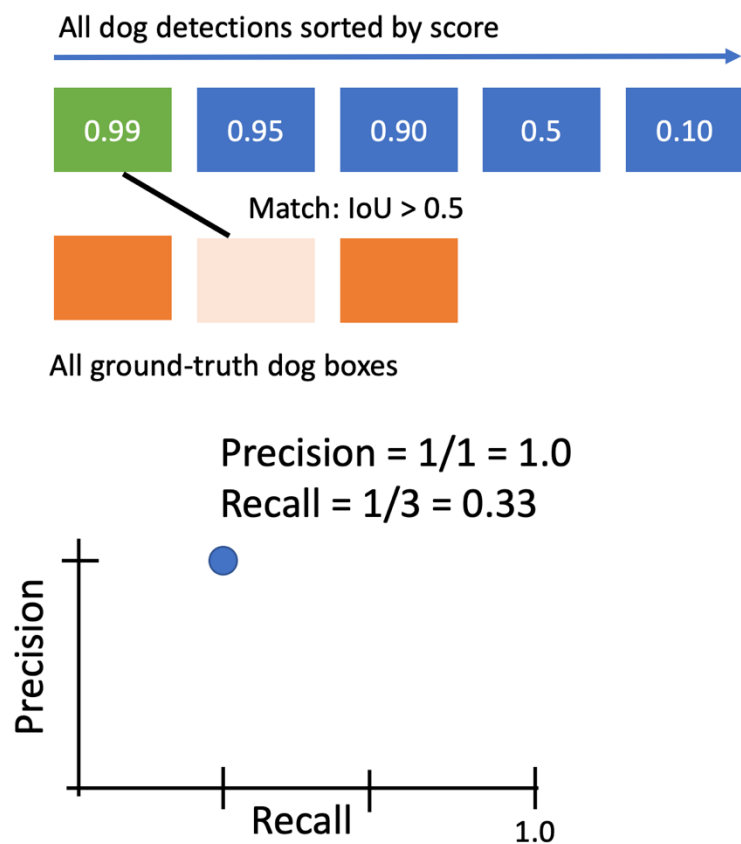
$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$



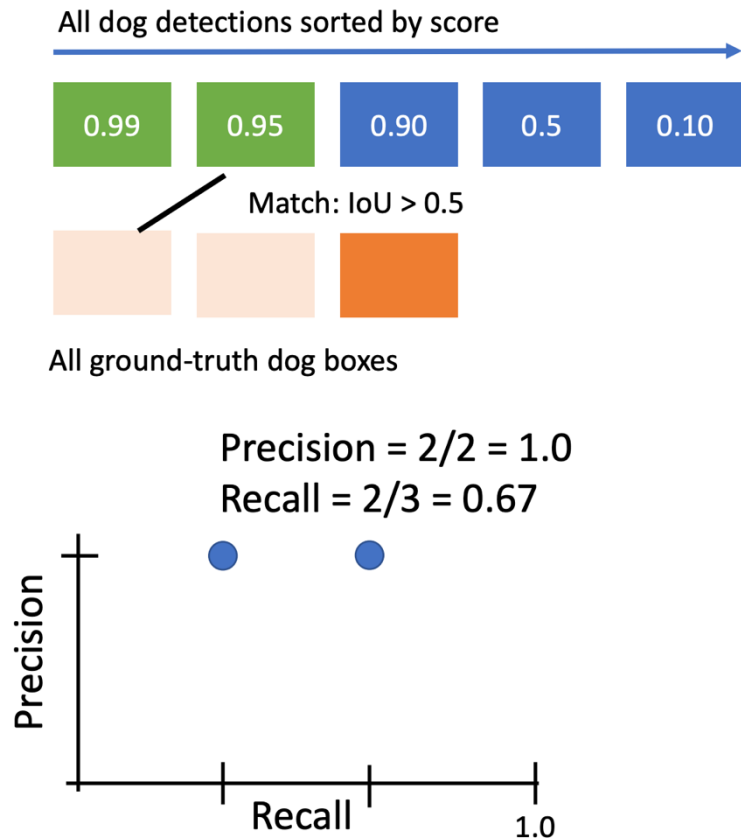
R-CNN INFERENCE

Solution) confidence threshold를 1에서 0까지 바꿔가며 precision/recall 계산
한 후 Precision-recall curve 그래프를 그림



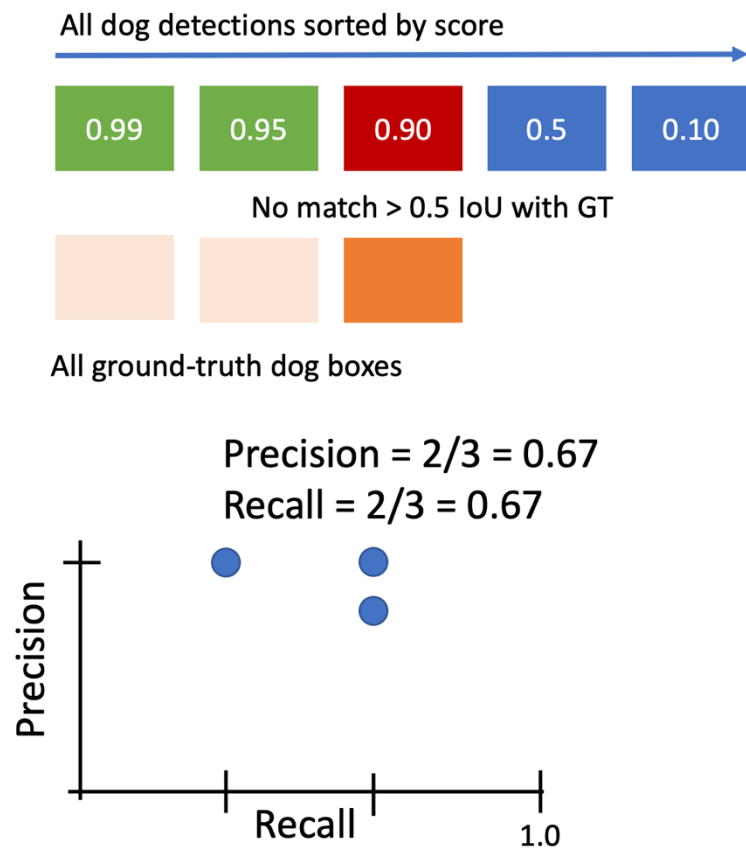
R-CNN INFERENCE

Solution) confidence threshold를 1에서 0까지 바꿔가며 precision/recall 계산
한 후 Precision-recall curve 그래프를 그림



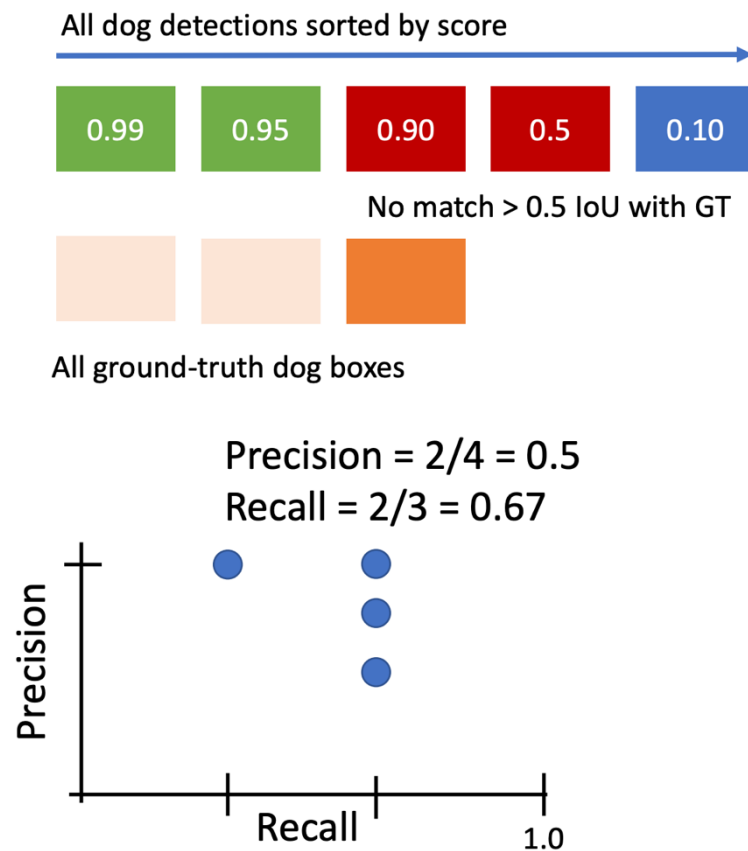
R-CNN INFERENCE

Solution) confidence threshold를 1에서 0까지 바꿔가며 precision/recall 계산
한 후 Precision-recall curve 그래프를 그림



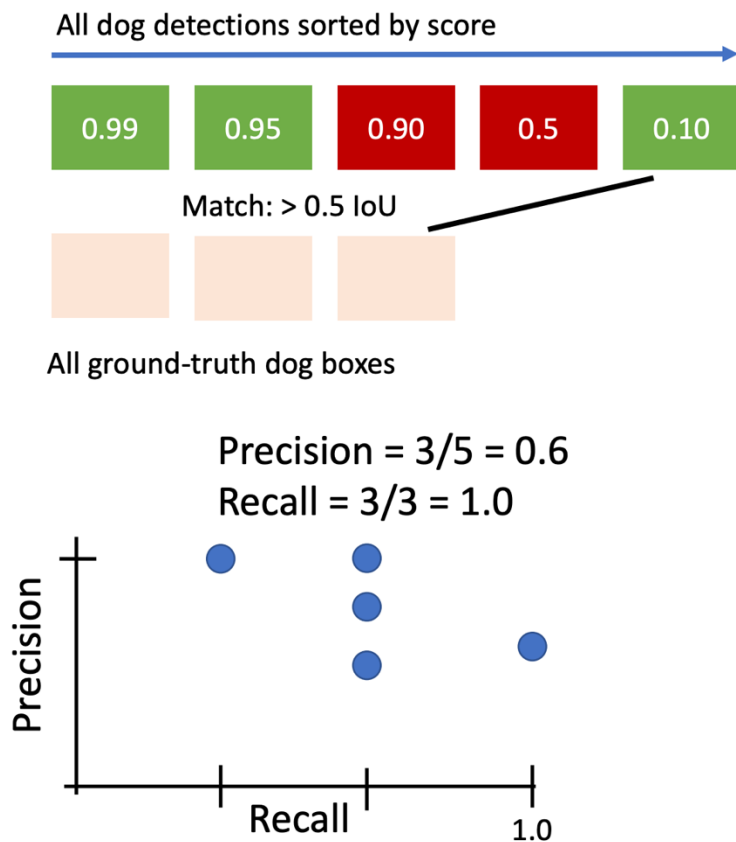
R-CNN INFERENCE

Solution) confidence threshold를 1에서 0까지 바꿔가며 precision/recall 계산
한 후 Precision-recall curve 그래프를 그림



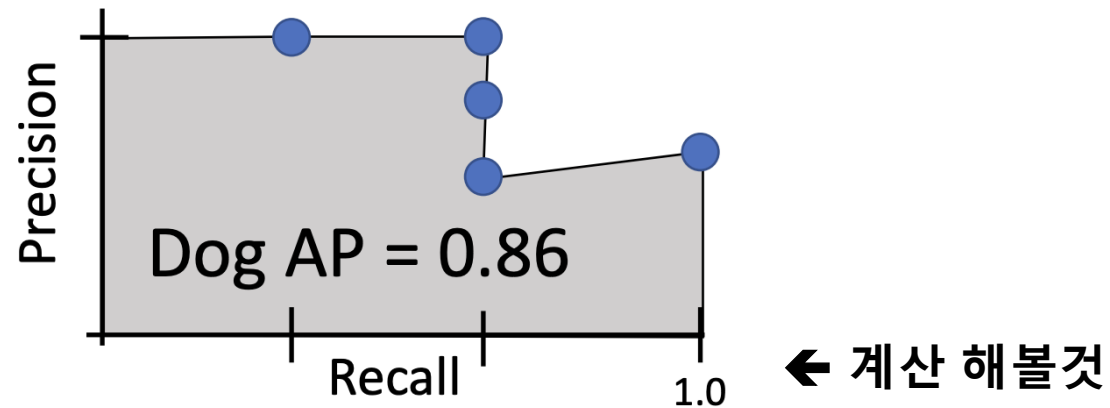
R-CNN INFERENCE

Solution) confidence threshold를 1에서 0까지 바꿔가며 precision/recall 계산
한 후 Precision-recall curve 그래프를 그림



R-CNN INFERENCE

- Average Precision (AP) : PR curve 아래의 면적
계산 편의를 위해 단조 감소 형태로 조정 후 계산하는 것이 일반적



- mean Average Precision (mAP) : 각 카테고리별 AP의 평균

Car AP = 0.65

Cat AP = 0.80

Dog AP = 0.86

mAP@0.5 = 0.77



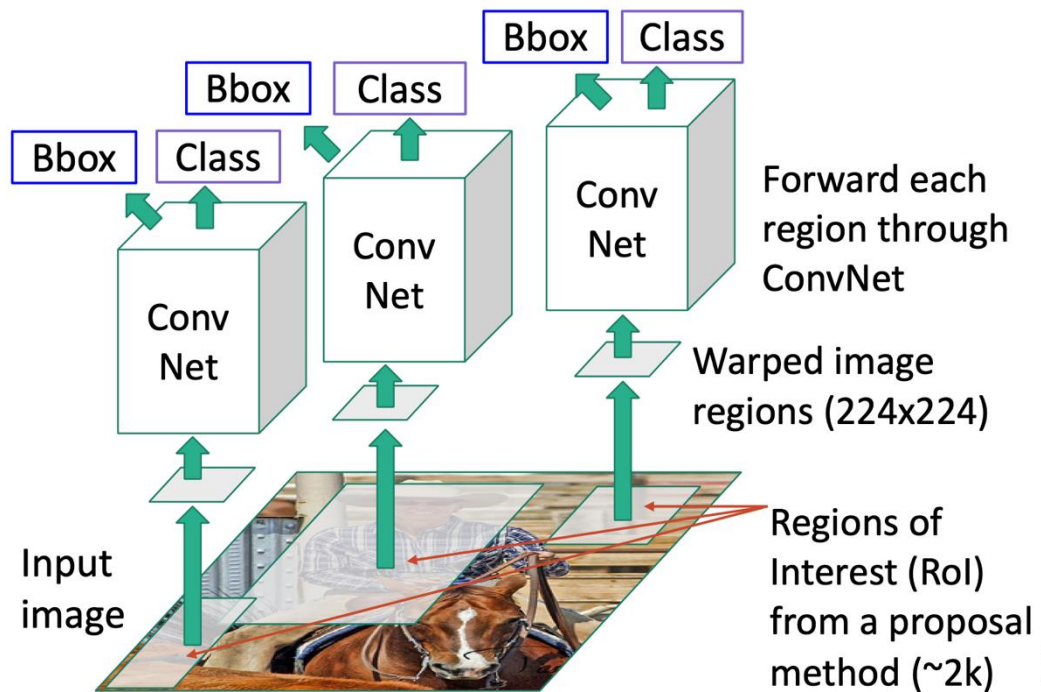


FAST R-CNN

The image features a minimalist design on a light gray background. Two thin, dark gray lines intersect: one line slopes downward from the top-left towards the bottom-right, and the other slopes more steeply downward from the top-center towards the bottom-right. To the right of the intersection, the text 'FAST R-CNN' is displayed in a bold, black, sans-serif font.

LIMITATION OF R-CNN

약 2천개에 해당하는 region proposal 각각에 대한 2천번의 forward pass 필요
→ 매우 느림

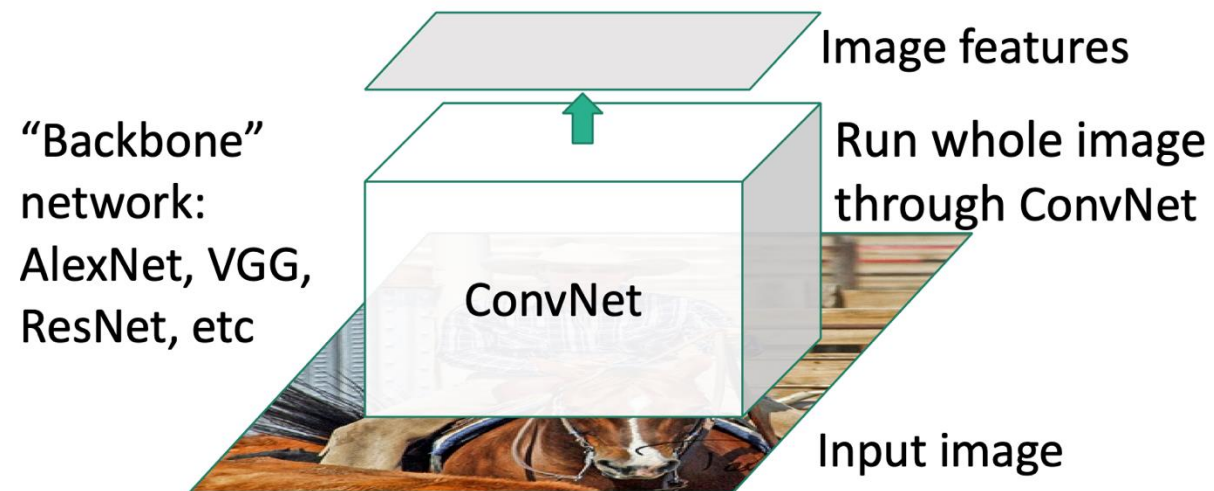


Girshick et al, "Rich feature hierarchies for accurate object detection and semantic segmentation", CVPR 2014.
Figure copyright Ross Girshick, 2015; [source](#). Reproduced with permission.

FAST R-CNN

ICCV 2015

Solution : image에 대한 forward pass는 한 번만 수행, feature map 공유



Girshick, "Fast R-CNN", ICCV 2015. Figure copyright Ross Girshick, 2015; [source](#). Reproduced with permission.



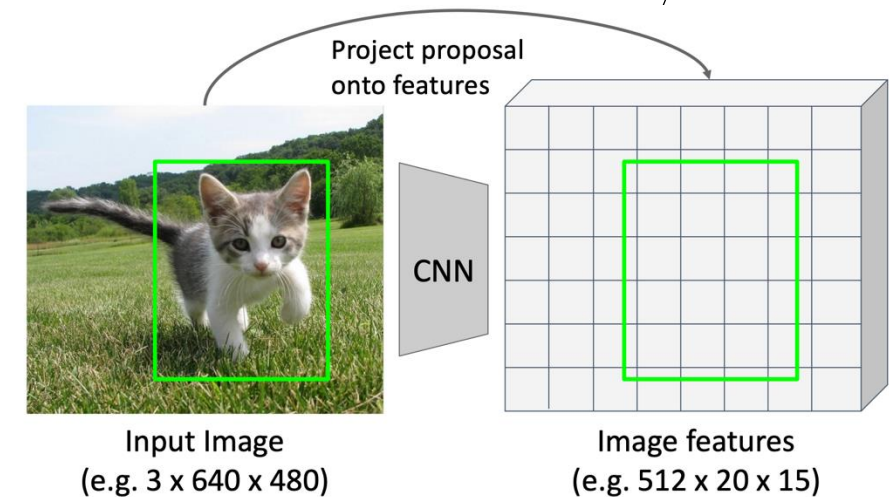
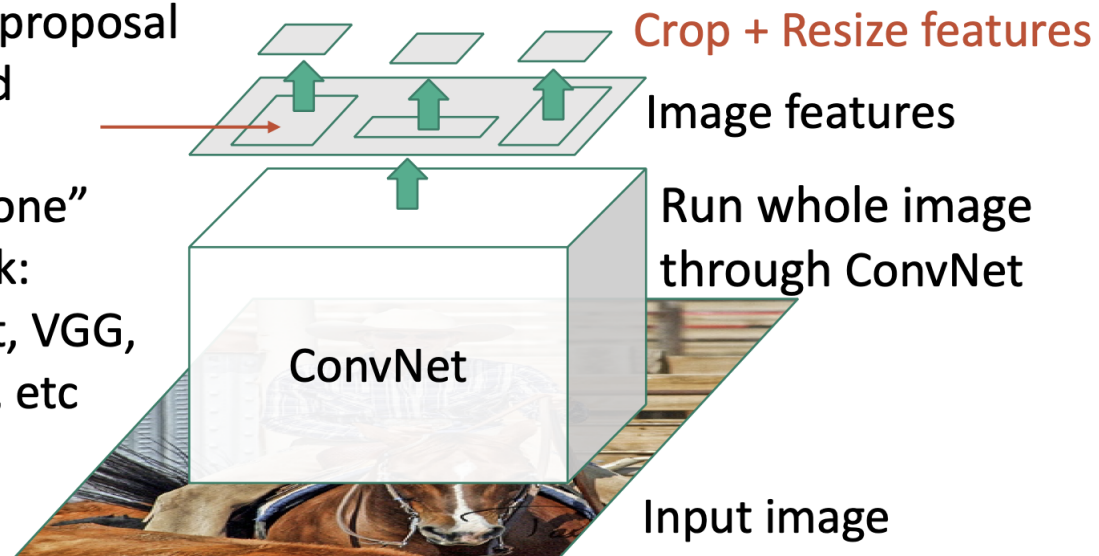
FAST R-CNN

ICCV 2015

Solution : image에 대한 forward pass는 한 번만 수행, feature map 공유

Regions of Interest (Rols)
from a proposal
method

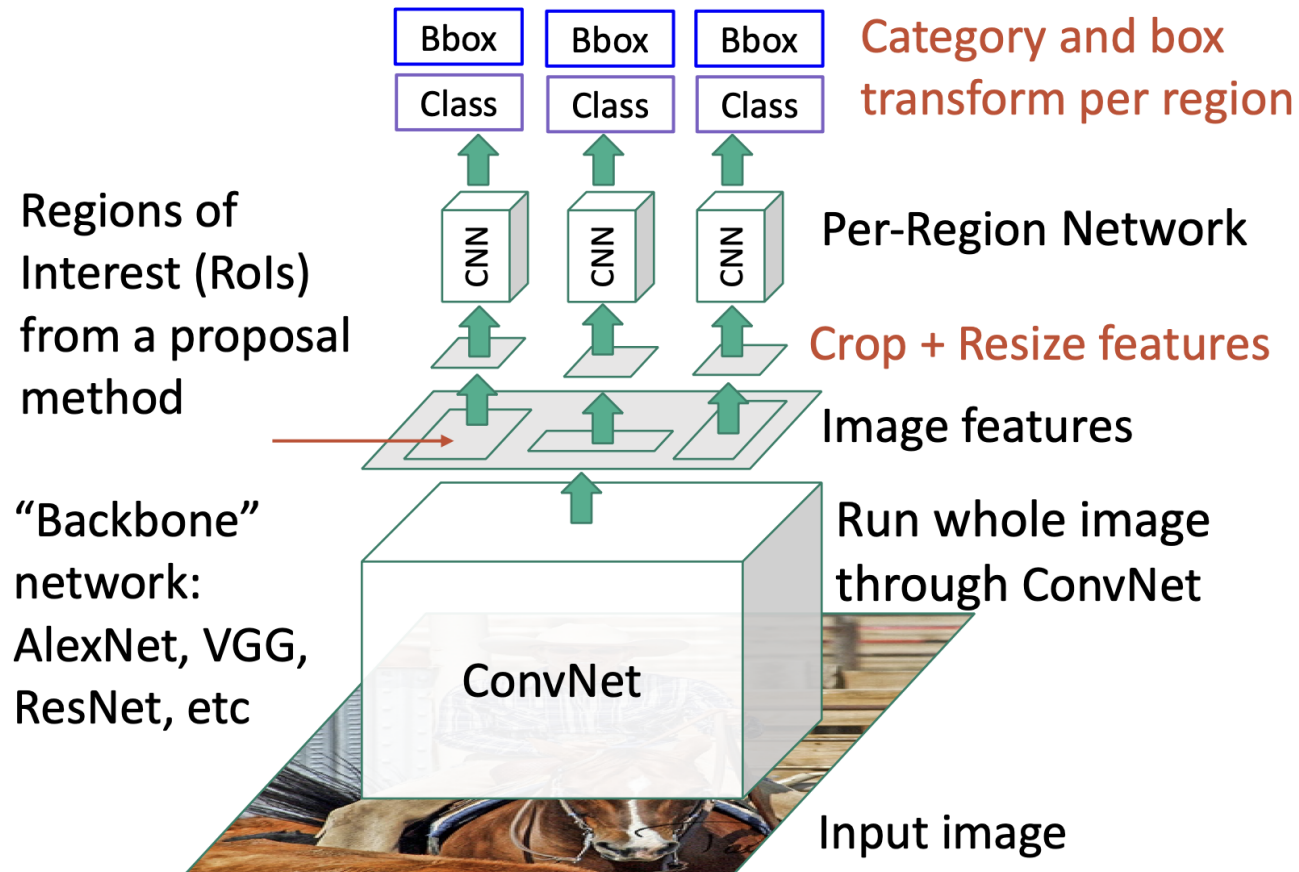
“Backbone”
network:
AlexNet, VGG,
ResNet, etc



FAST R-CNN

ICCV 2015

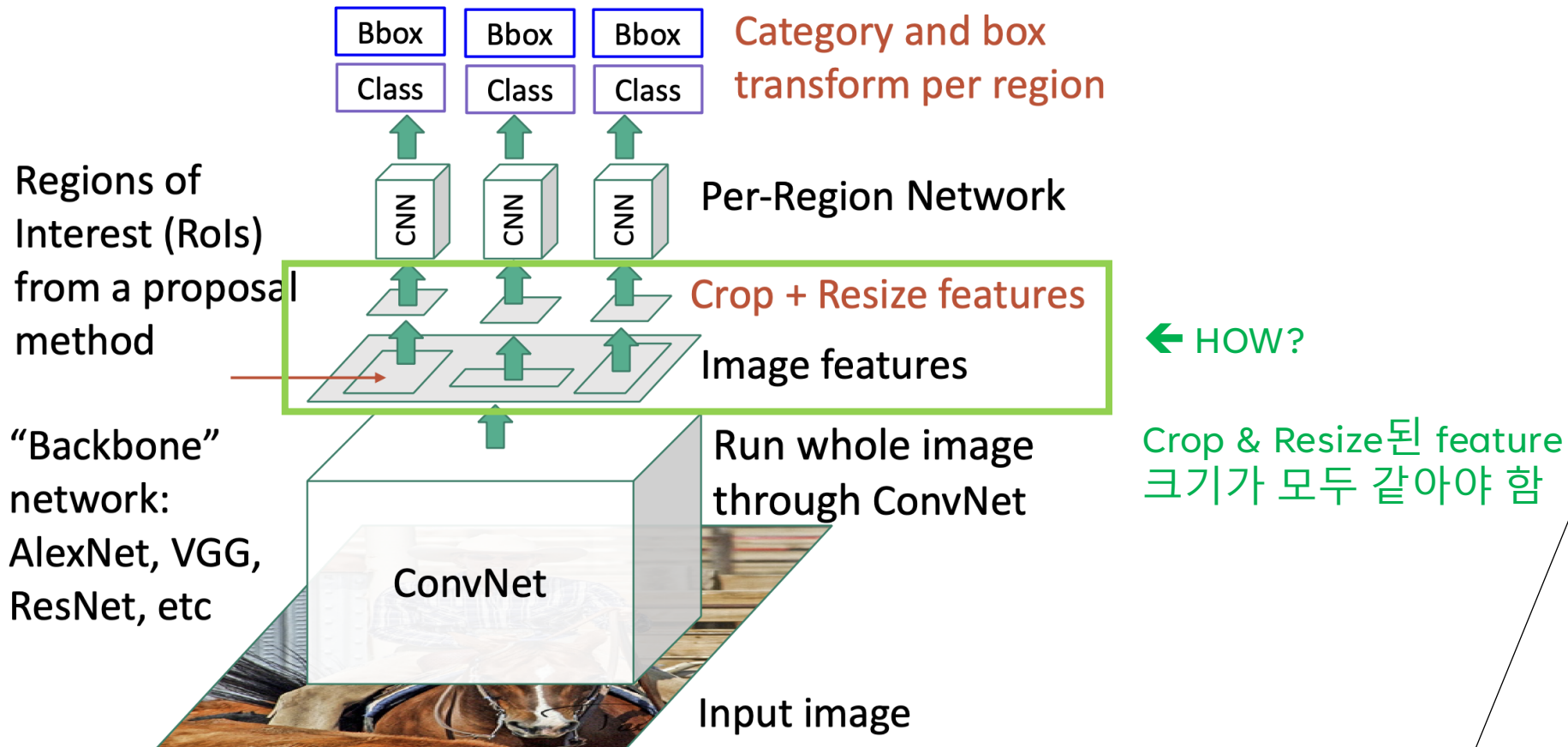
Solution : image에 대한 forward pass는 한 번만 수행, feature map 공유



FAST R-CNN

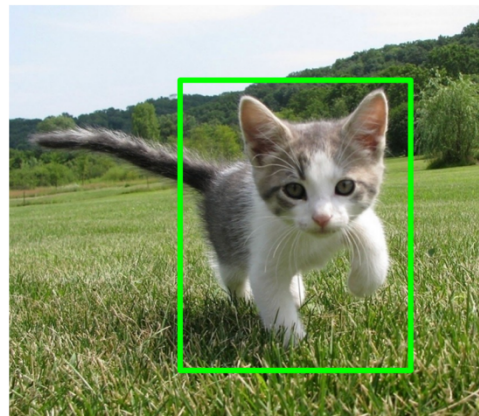
ICCV 2015

Solution : image에 대한 forward pass는 한 번만 수행, feature map 공유



ROI POOLING

Cropping Features: RoI Pool



Input Image
(e.g. 3 x 640 x 480)

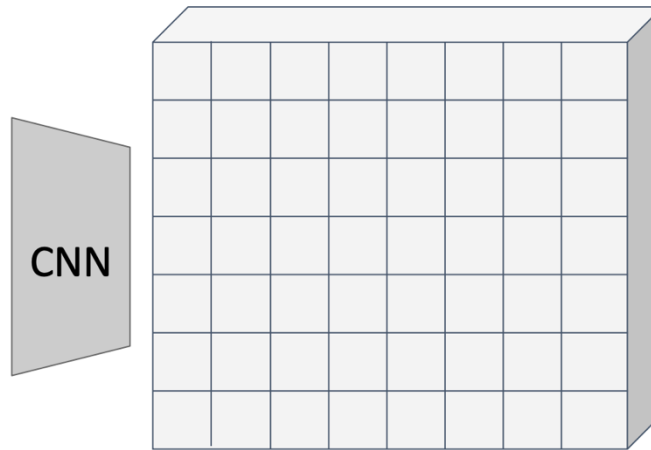


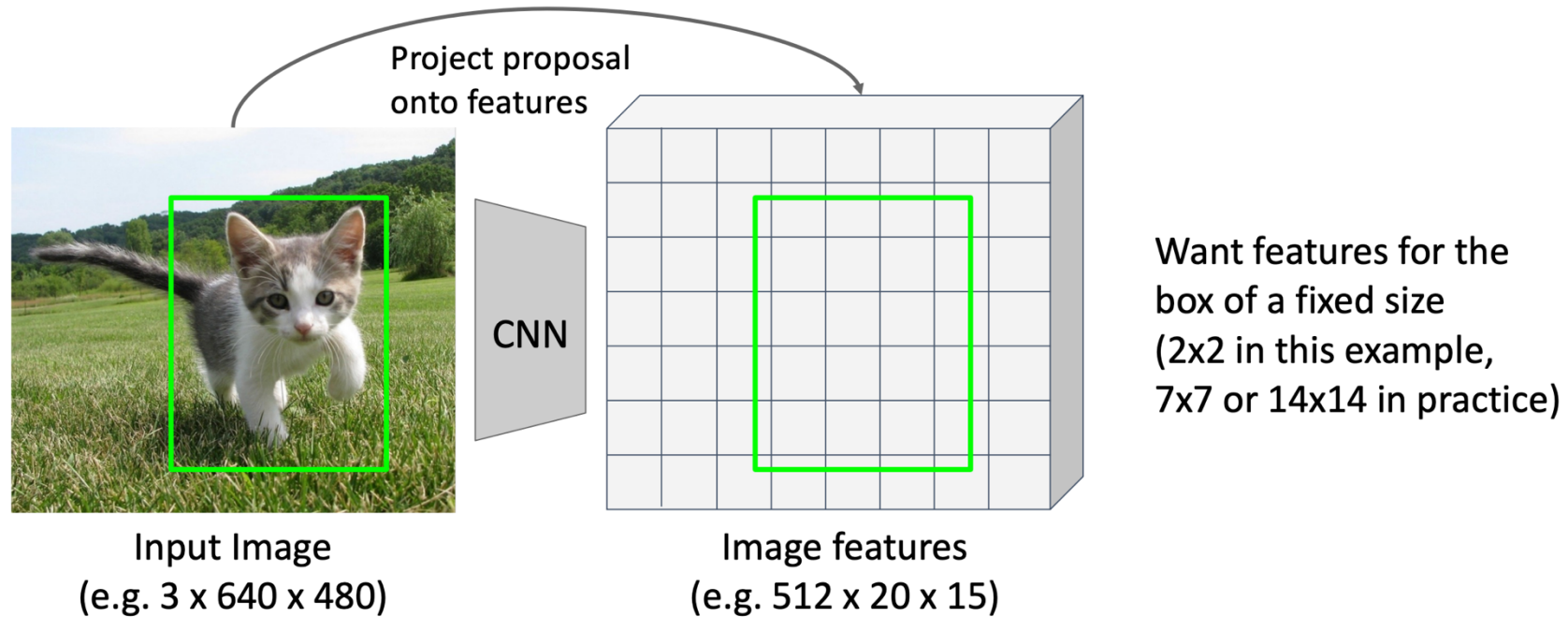
Image features
(e.g. 512 x 20 x 15)

Want features for the
box of a fixed size
(2x2 in this example,
7x7 or 14x14 in practice)



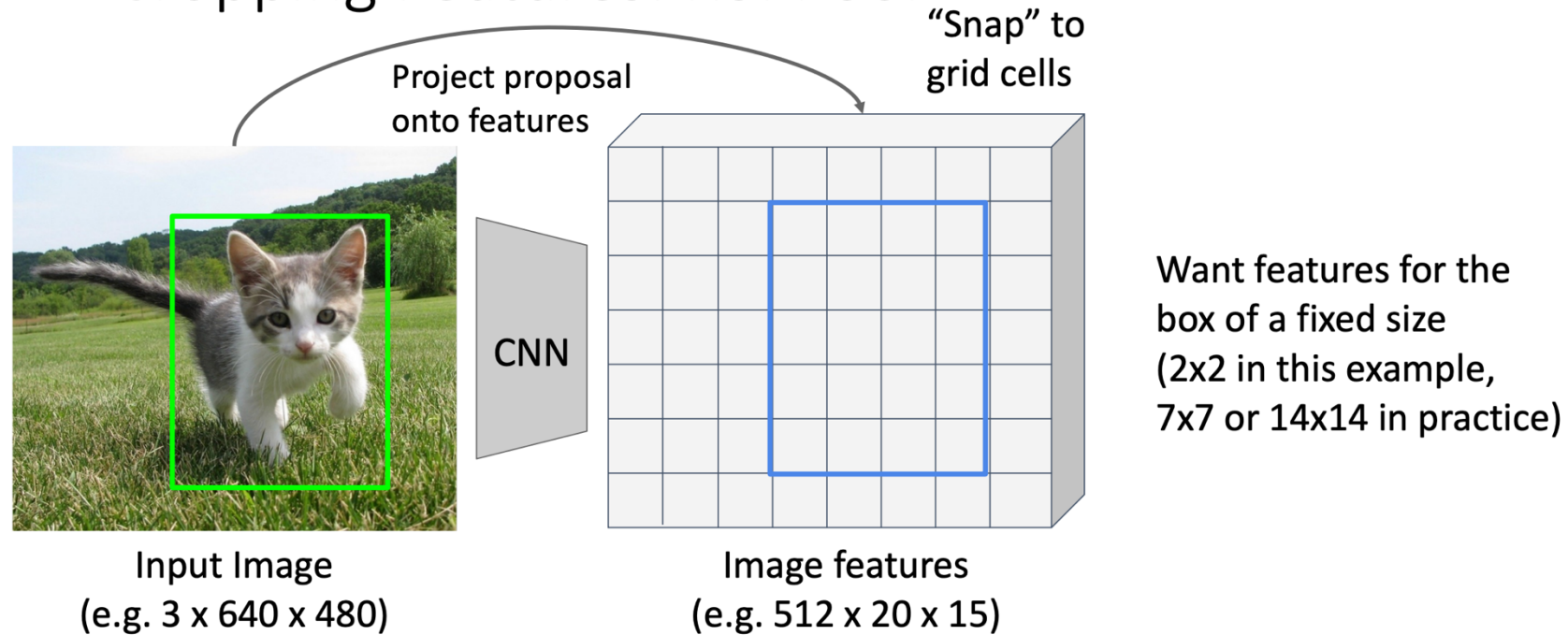
ROI POOLING

Cropping Features: RoI Pool



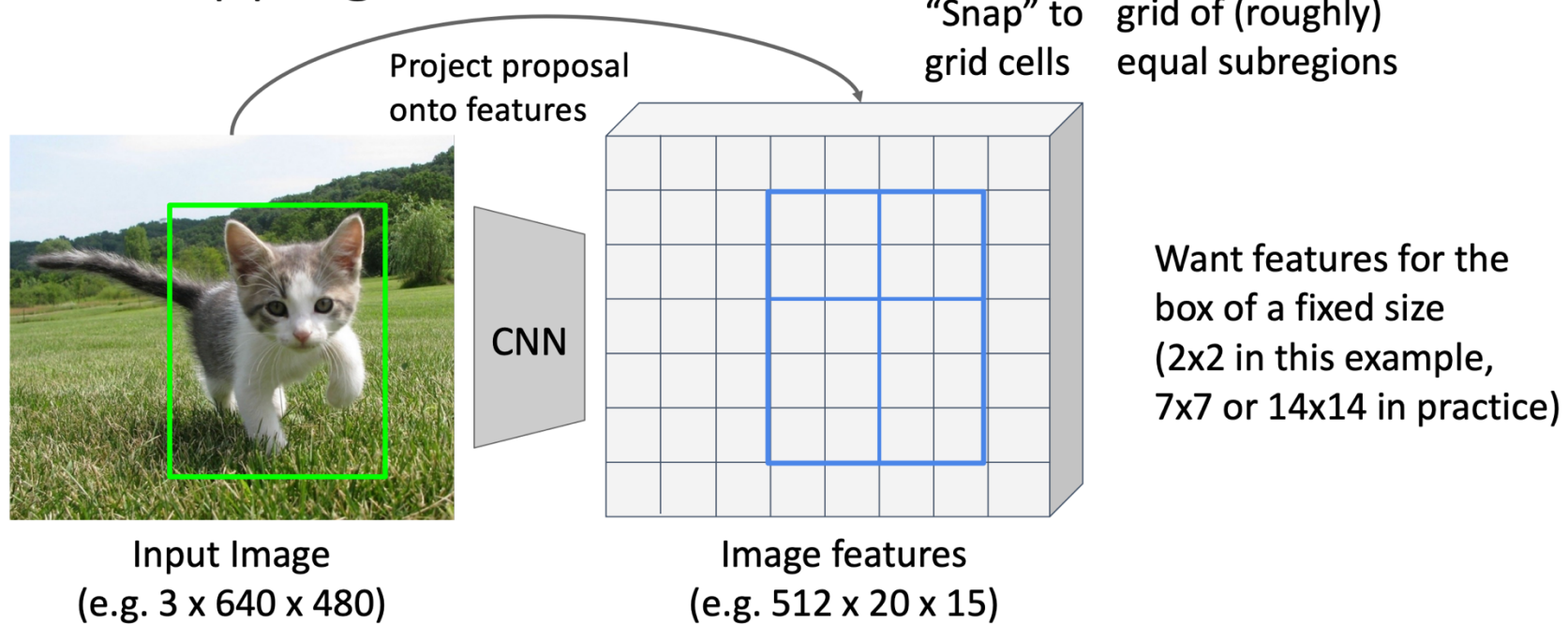
ROI POOLING

Cropping Features: RoI Pool



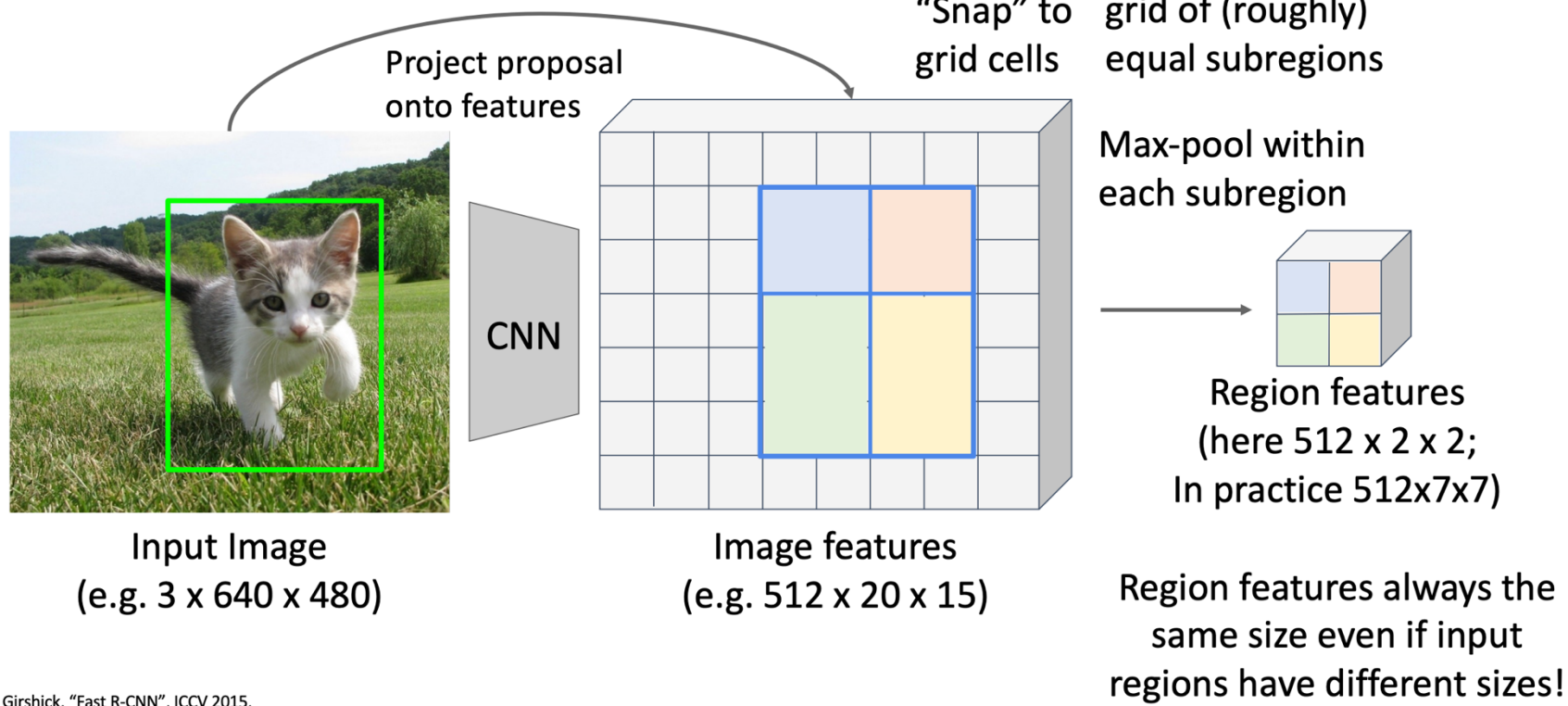
ROI POOLING

Cropping Features: RoI Pool



ROI POOLING

Cropping Features: RoI Pool



Girshick, "Fast R-CNN", ICCV 2015.



Two thin, dark gray lines intersect diagonally on a light gray background. One line slopes downward from left to right, and the other slopes downward from right to left, crossing the first line.

SUMMARY

SUMMARY

1. R-CNN inference

2. Fast R-CNN



NEXT LECTURE

Perception (4) – 2D object detection – Continued

Faster R-CNN



Two thin, dark gray lines intersect diagonally on a light gray background. One line runs from the top-left towards the bottom-right, and the other runs from the top-right towards the bottom-left. They cross each other in the upper-left quadrant of the image.

QUESTIONS?